



AI-POWERED VEHICLE DAMAGE DETECTION

Insurance Claims

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ITAI 1378 – Computer Vision and AI
GitHub Midterm

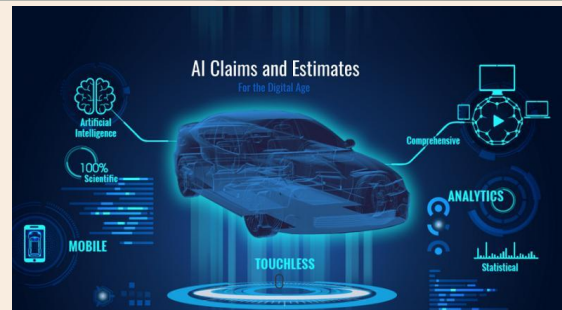


PROBLEM:

- ✓ Insurance companies rely on manual inspections for vehicle damage.
- ✓ This process is slow, costly, and inconsistent.
- ✓ Customers wait days for repair estimates.

SOLUTION:

- ✓ Users upload a photo of a damaged vehicle.
- ✓ AI model detects damage areas (dents, scratches, cracks).
- ✓ System classifies the type of damage.
- ✓ Generates an estimated repair cost automatically.



TECHNICAL APPROACH:

- ✓ Technique: Object Detection
- ✓ Model: YOLOv8
- ✓ Frameworks:
 - PyTorch
 - Ultralytics YOLO
 - OpenCV
- ✓ Justification:
YOLOv8 is fast and accurate, ideal for real-time damage detection.

DATA PLAN:



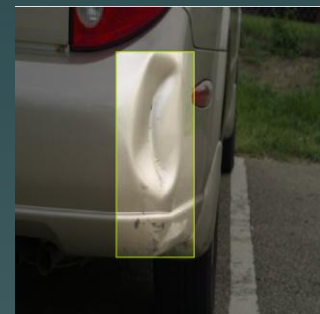
Dataset:

- Roboflow Universe – Automobile Damage Detection
- ~6,900 labeled images
- Classes: dent, scratch, crack



Preparation:

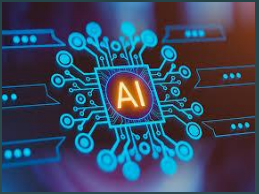
- Export in YOLO format
- Split into train/validation/test
- Optional data augmentation



PIPELINE:



✓ Vehicle Image



✓ Image Preprocessing



✓ Damage Detection (YOLOv8)



✓ Damage Classification



✓ Repair Cost Estimation



✓ Output Report



METRICS:



Primary Metric:

- Detection accuracy $\geq 85\%$



Secondary Metrics:

- Damage classification accuracy $\geq 80\%$
- Inference time < 1 second per image

WEEK-BY-WEEK PLAN:

Week	Task	Milestone
Week 10	Dataset collection	Dataset ready
Week 11	Train object detection model	Model running
Week 12	Improve accuracy	Stable performance
Week 13	Add cost estimation module	Demo ready
Week 14	Testing & documentation	Final version
Week 15	Presentation	Project demonstration

CHALLENGES & BACKUP PLANS:

Challenge	Plan B
Dataset too small	Use alternative Roboflow datasets
Low model accuracy	Apply data augmentation or fine-tune larger YOLO model
Training too slow	Use Google Colab GPU

RESOURCES NEEDED:

- ✓ Compute: Google Colab GPU
- ✓ Frameworks:
 - PyTorch
 - Ultralytics YOLO
 - OpenCV
- ✓ Estimated Cost: \$0
- ✓ Optional Demo: Streamlit web app